

Gebze Technical University
Dep. of Mechanical Engineering

System Dynamics
and Vibrations (ME 331)

Lecture 1: Introduction and Fundamentals

Saeed Lotfan

Course introduction

Thursday 13:00–16:00 • Harita Z06

Lectures and tablet derivations

Weekly recitation practice and assigned homework

Midterm 25% Quizzes 10% Homework 10%

Project 30% Final 25%

Today: physical system, coordinates, elements, equation

Lecture overview

1.1 Why study vibration? A brief history

1.2 Basic concepts of vibration

1.3 Degrees of freedom and idealization

1.4 Classification of vibration

1.5 Vibration analysis procedure

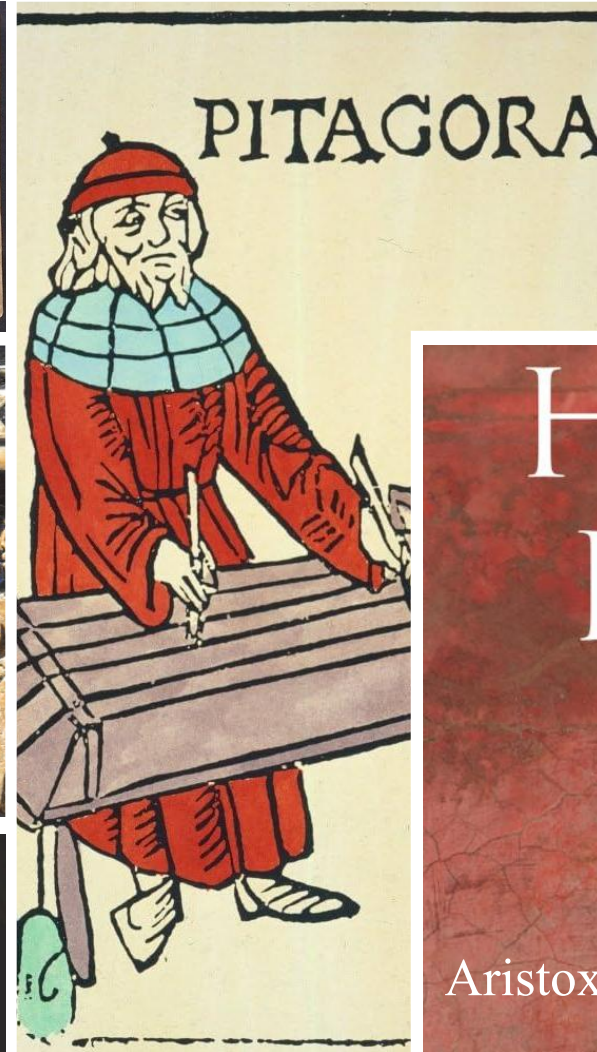
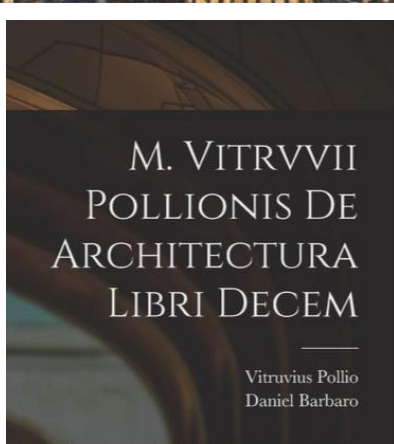
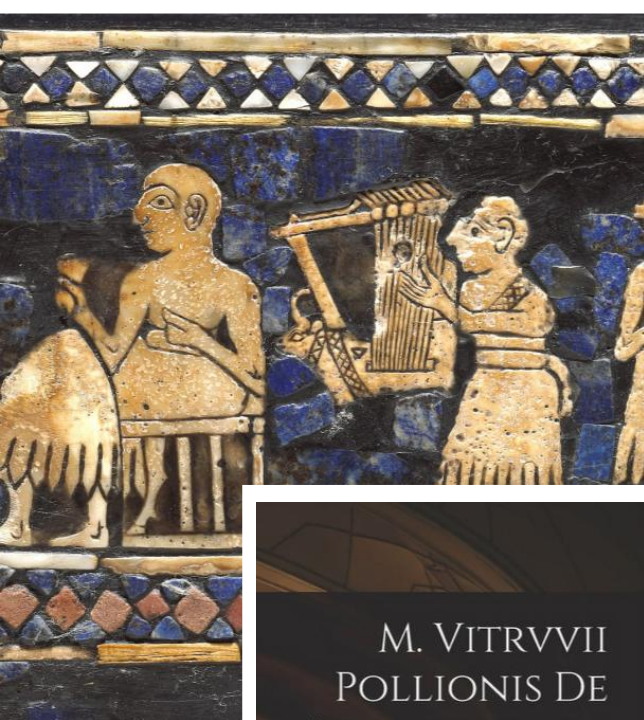
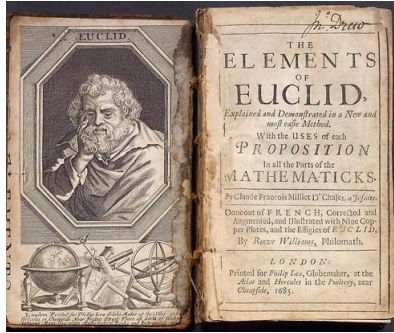
1.6 Spring elements

1.7 Mass and inertia elements

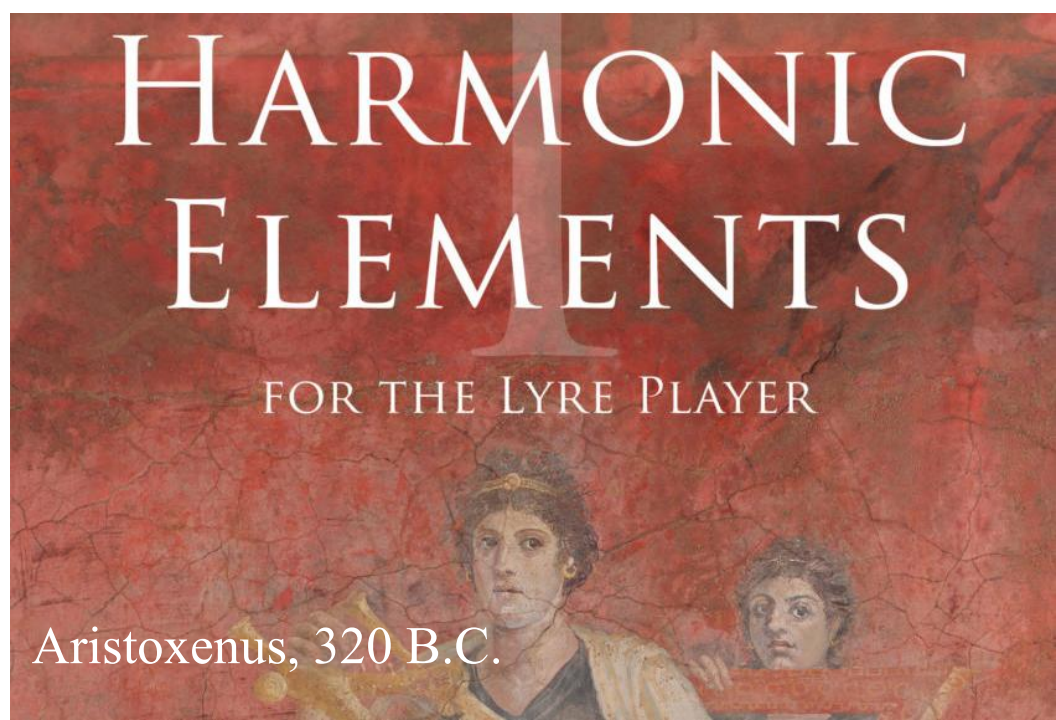
1.8 Damping elements

1.9 Physical system to vibration model

300 B.C.,
*Introduction
to Harmonics*



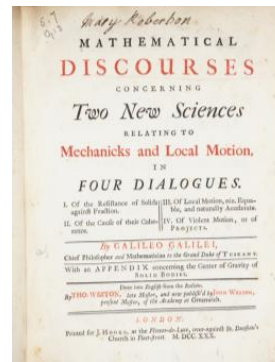
Aristotle, c. 350 B.C.
Early writings on sound and music



1.1 History of vibration

1.1 History of vibration

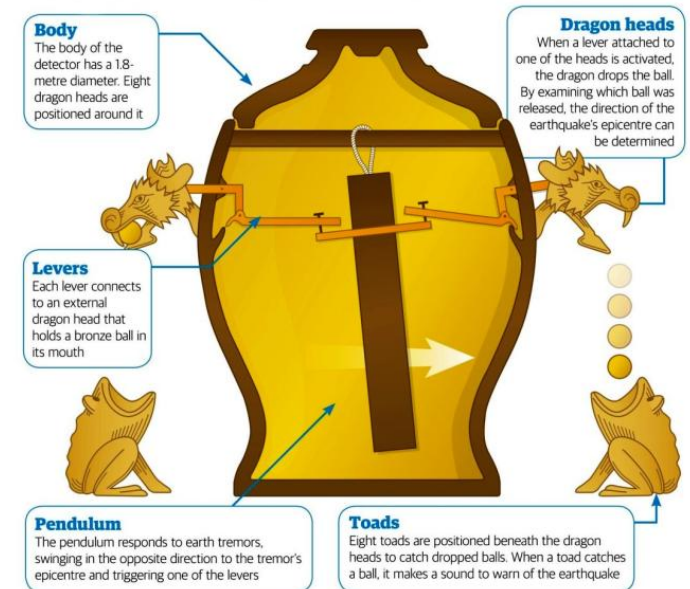
Galileo Galilei (1564–1642)



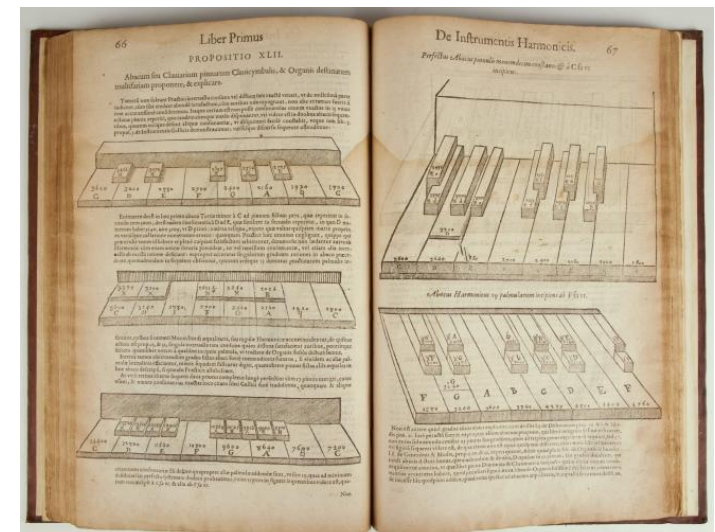
Discourses Concerning Two New Sciences



Marin Mersenne (1588–1648),
Father of Acoustics



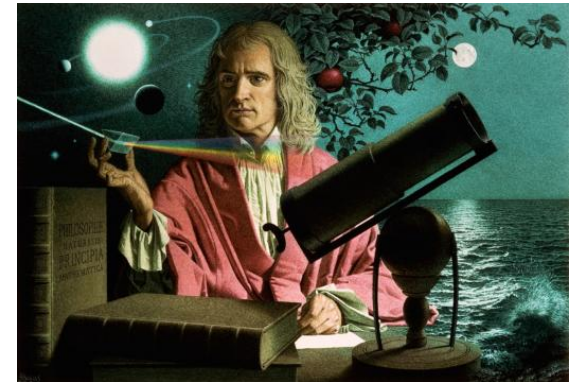
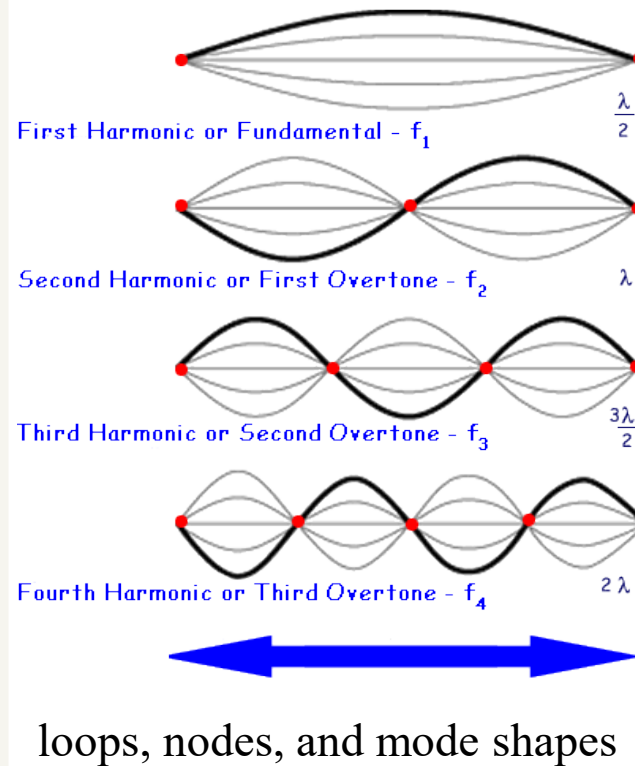
A.D. 132. Zhang Heng



Harmonicorum Liber



Joseph Sauveur (1653–1716)
Coined the word “acoustics”:

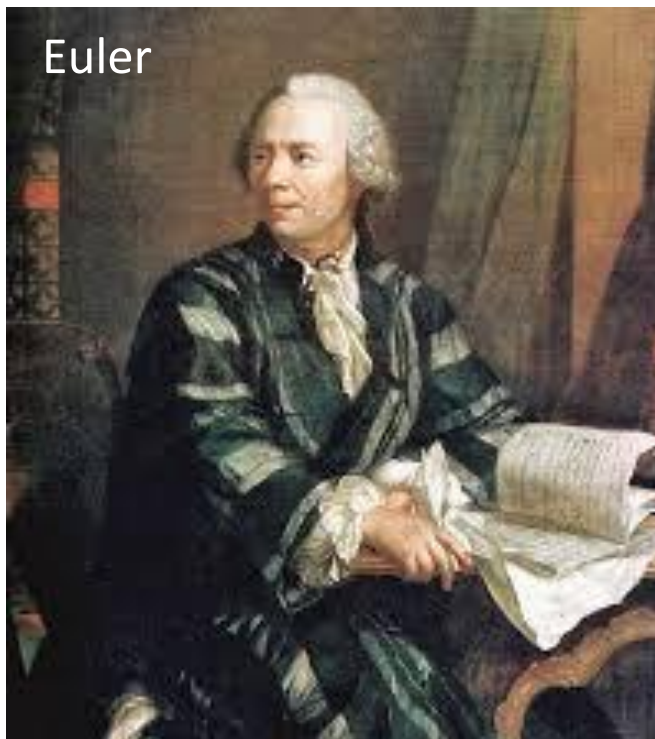
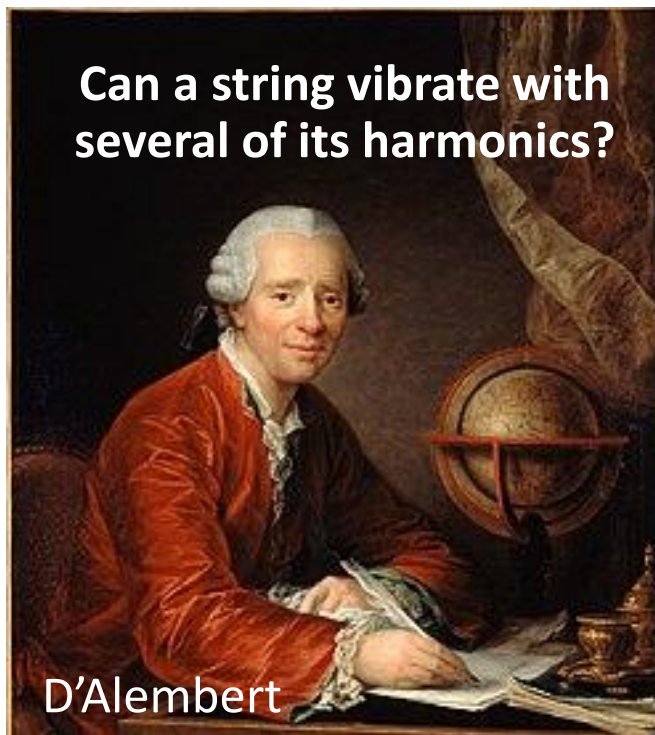
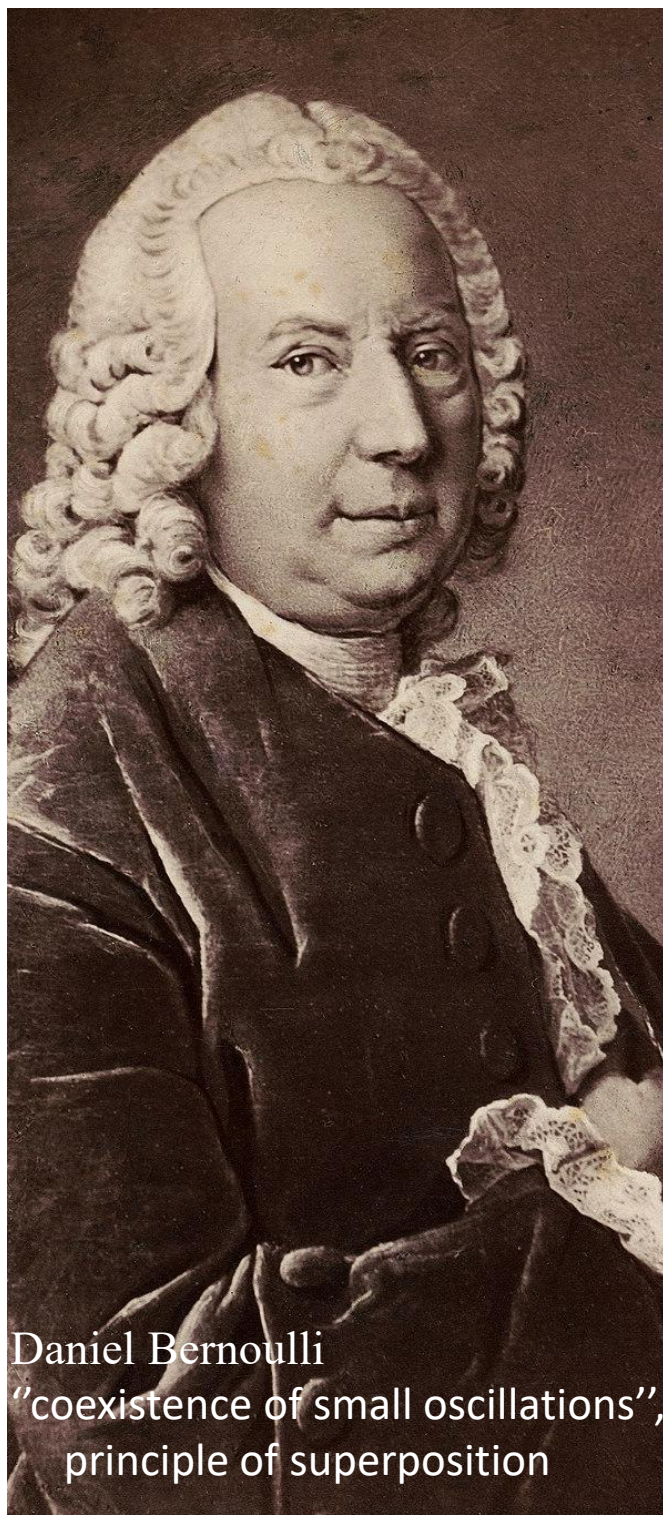


Sir Isaac Newton (1642–1727)
Philosophiae Naturalis
Principia Mathematica

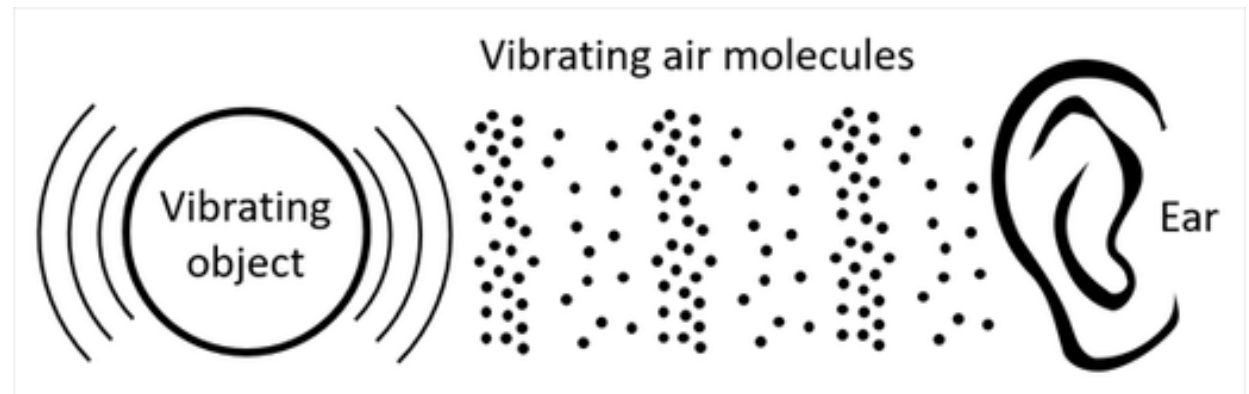


Brook Taylor (1685–1731)
String vibration equations
and experimental agreement

1.1 History of vibration



1.1 Why study vibration?



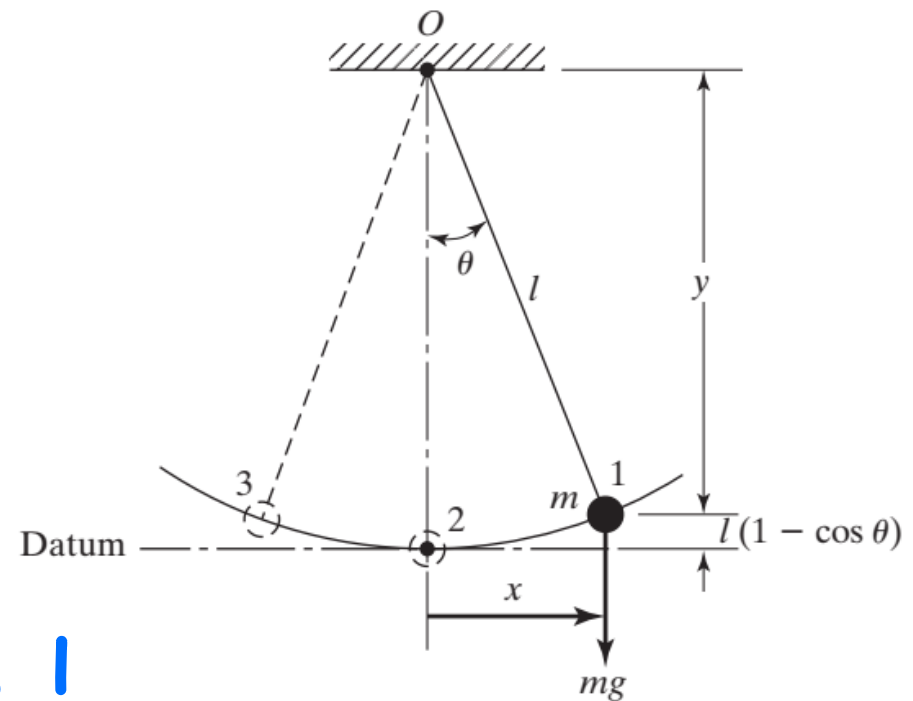
1.2 Basic concepts of vibration

Equilibrium • Excitation and response • Energy exchange

$$T = \frac{1}{2} m l^2 \dot{\theta}^2$$

$$U = mgl(1 - \cos \theta)$$

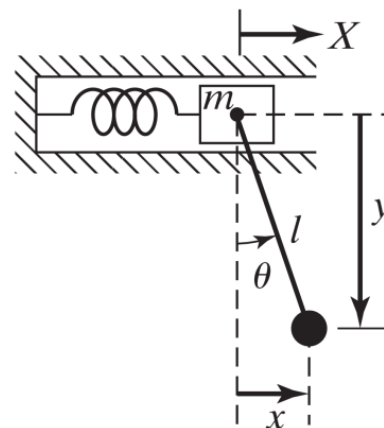
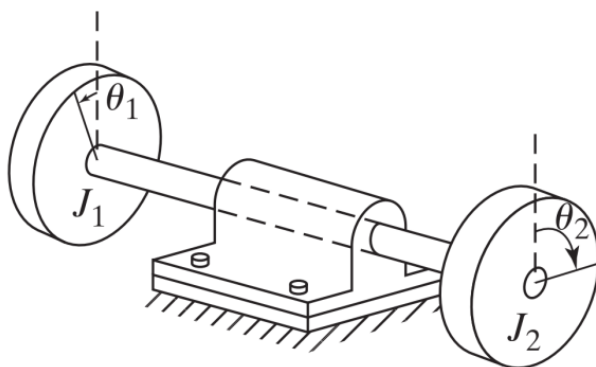
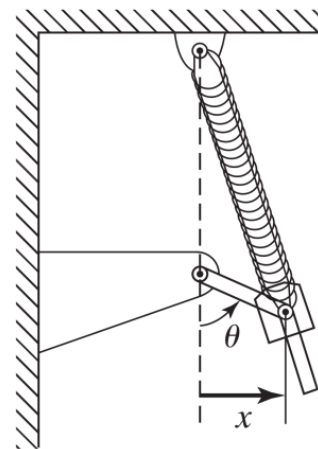
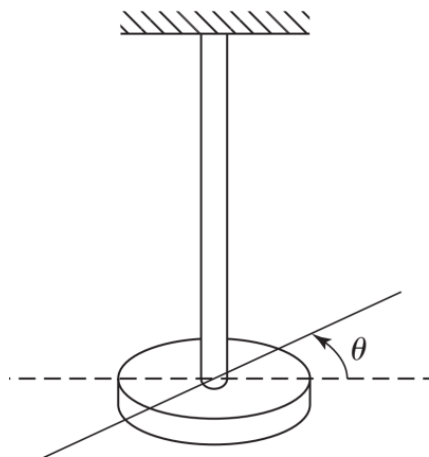
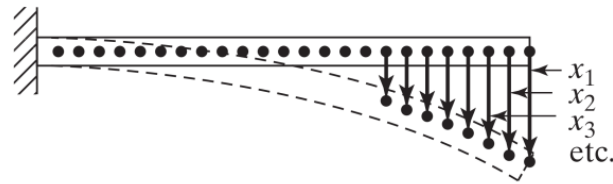
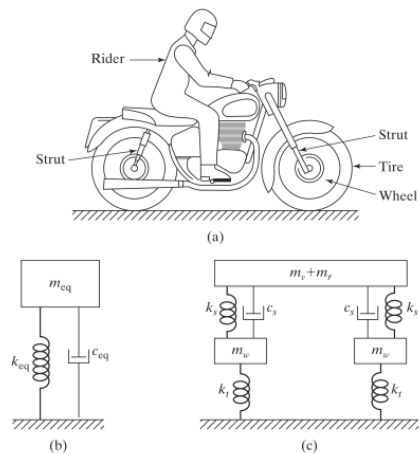
non
+ restoring force!



1.3 Degrees of freedom

Independent coordinates
Constraints

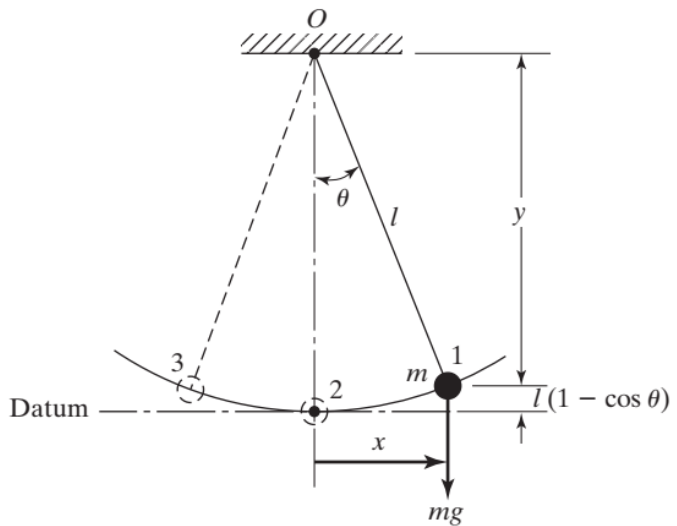
Discrete and continuous models



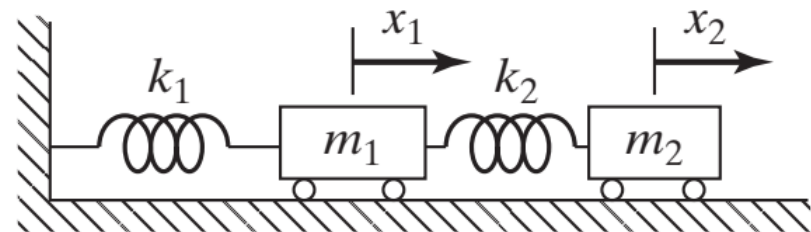
1.3 Degrees of freedom

Independent coordinates and constraints

Fixed-length pendulum



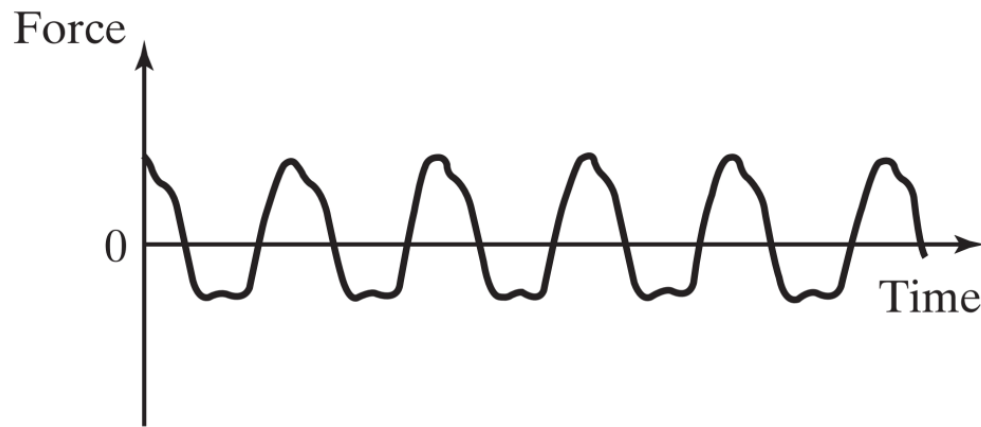
Two translating masses



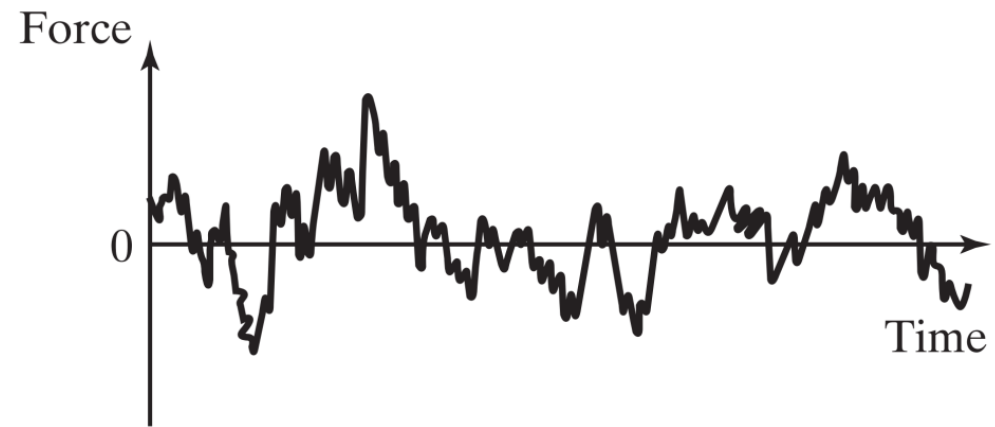
How does a continuous beam differ from these models?

1.4 Classification of vibration:

Free/Forced, Damped/Undamped,
Linear/Nonlinear, Deterministic/Random



(a) A deterministic (periodic) excitation



(b) A random excitation

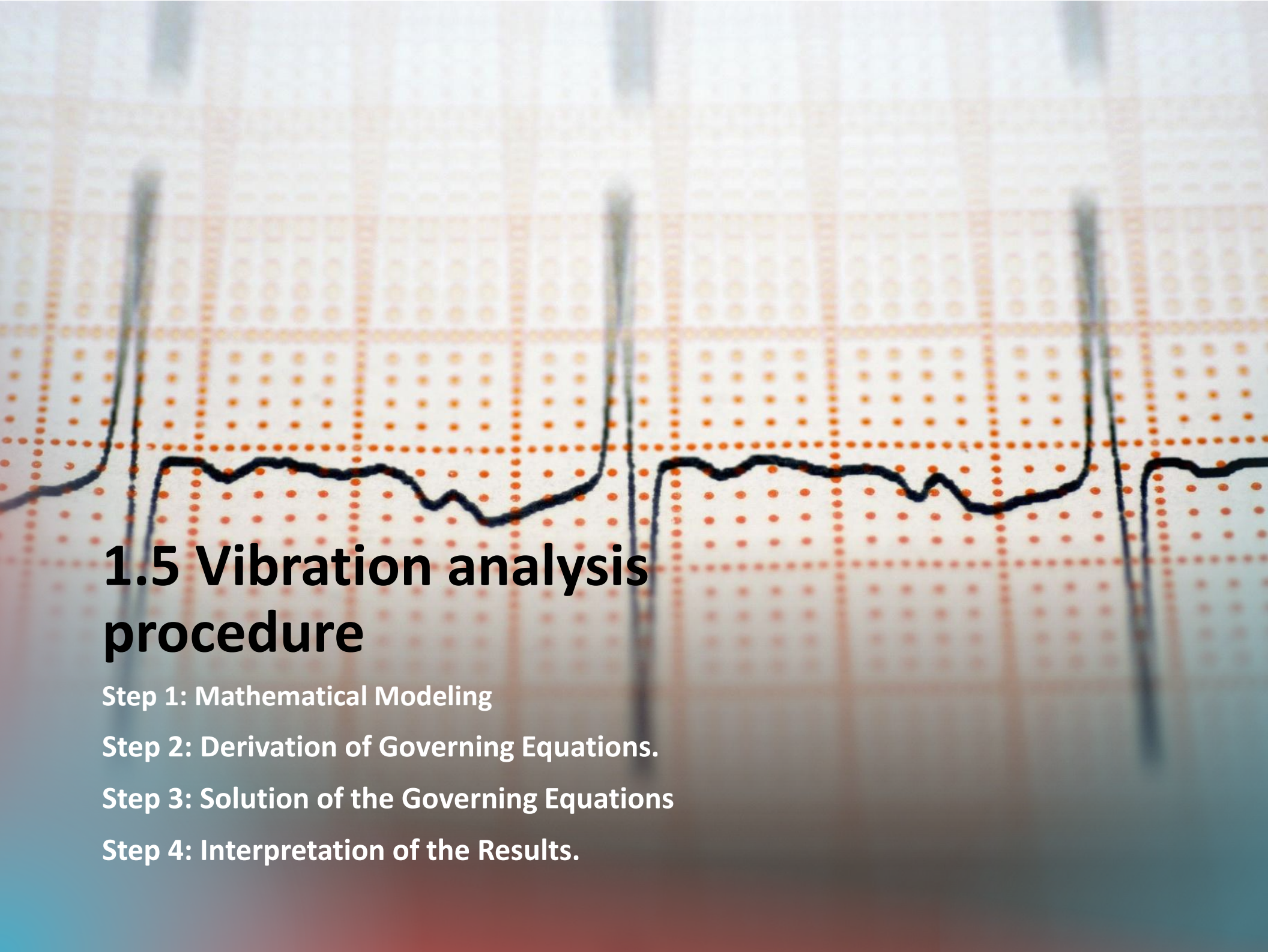
1.4 Classification: examples

Free or forced? Damped or undamped? Linear or nonlinear?

A ruler released after a small deflection

A machine under a prescribed sinusoidal force

A vehicle travelling over a rough road



1.5 Vibration analysis procedure

Step 1: Mathematical Modeling

Step 2: Derivation of Governing Equations.

Step 3: Solution of the Governing Equations

Step 4: Interpretation of the Results.

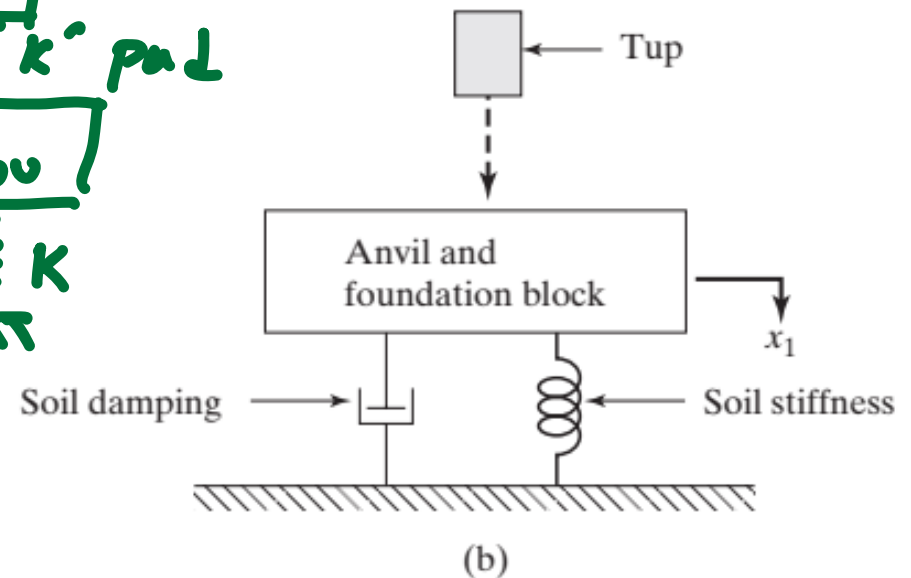
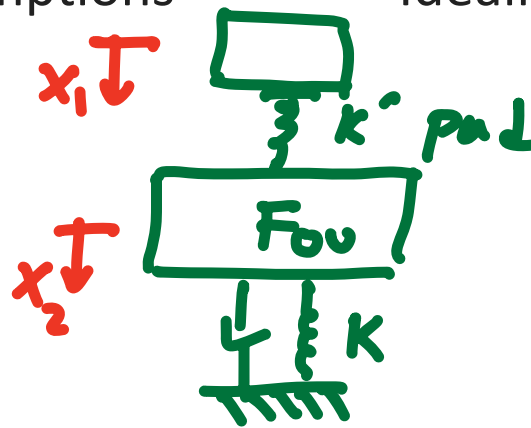
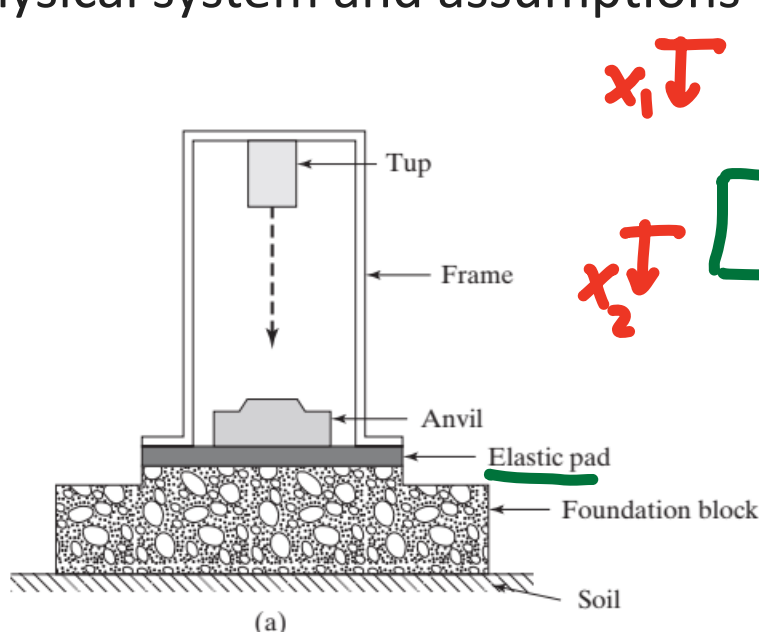
1.5 A model for an engineering question

Machine on elastic supports: what motion matters?

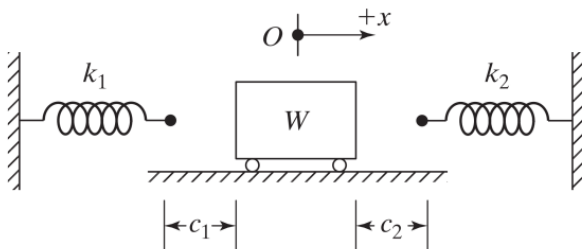
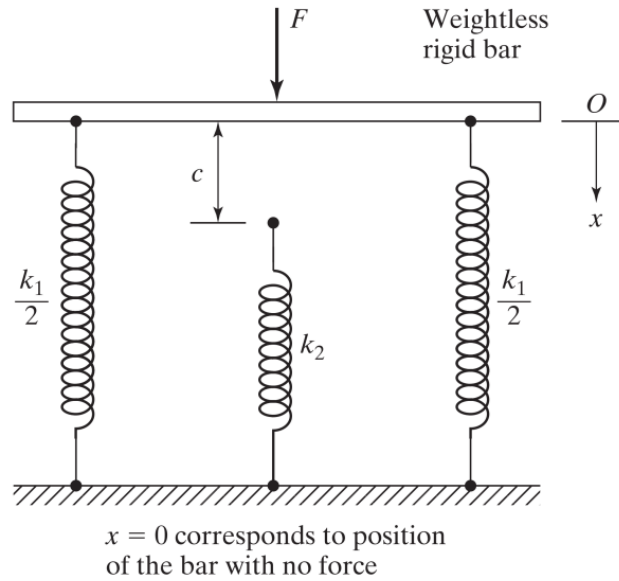
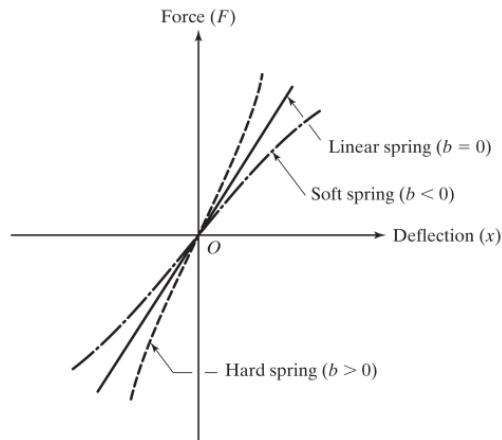
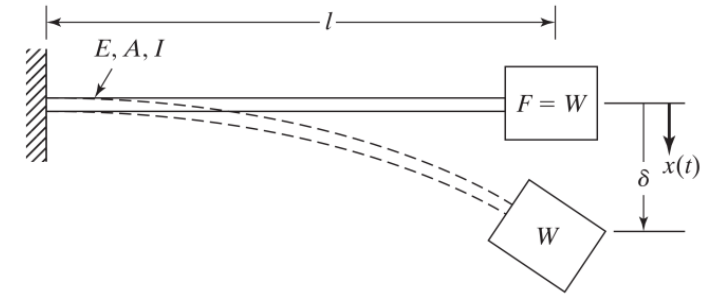
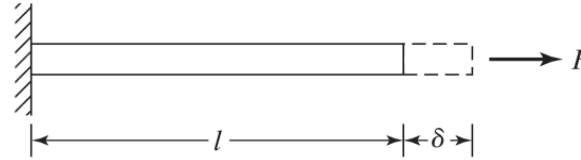
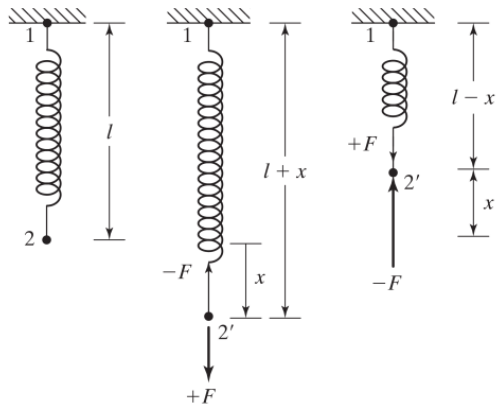


Physical system and assumptions

Idealized model and response



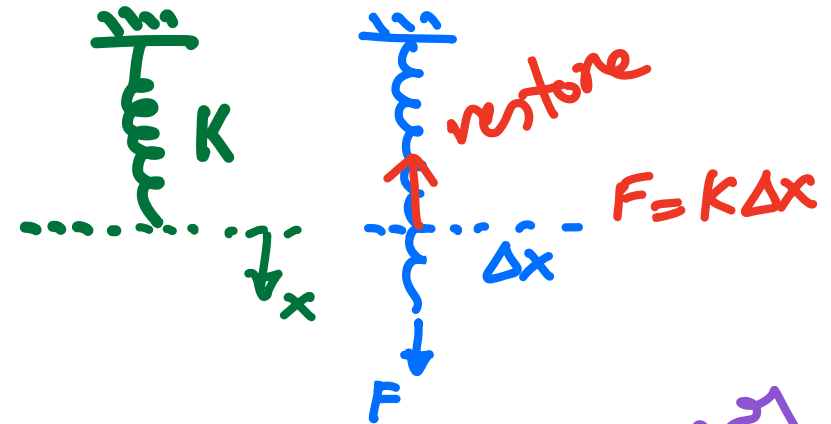
How would we check whether the model is adequate?



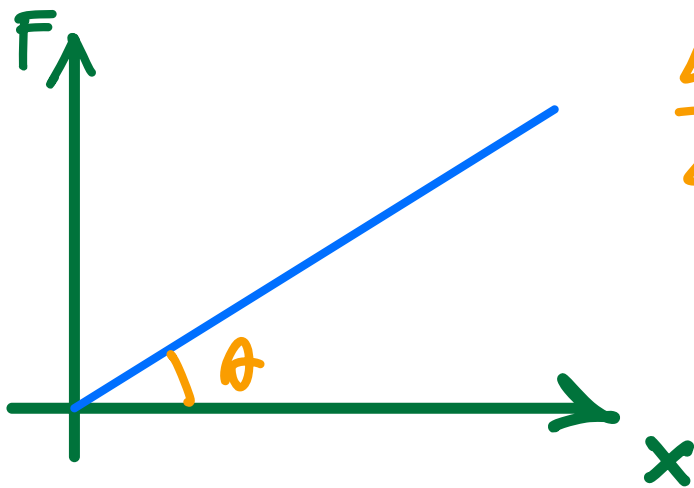
1.6 Spring elements

1.6 Spring force and deformation

Restoring force • Stiffness • Stored energy



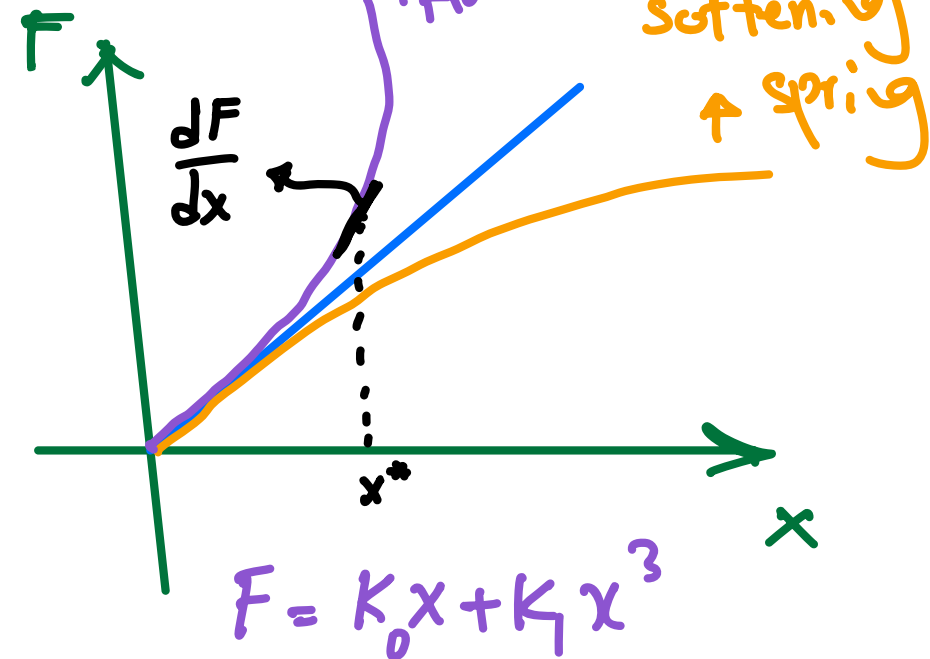
Linear spring



$$\frac{\Delta F}{\Delta x} = K$$

$$\tan \theta = K$$

Nonlinear spring

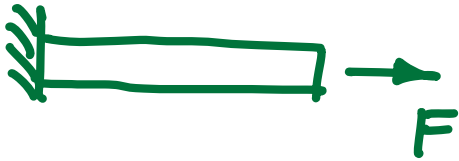


$$\left. \frac{dF}{dx} = K_0 + 3K_1 x^2 \right|_{x=x^*} = K_{eq}$$

1.6 Elastic members as springs

Geometry, material properties and boundary conditions

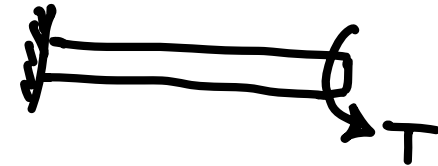
Axial rod



Cantilever tip



Shaft in torsion



Which assumptions make each stiffness formula valid?

1.6 Stiffness: a quick check

Uniform axial rod under a 1 kN tensile load

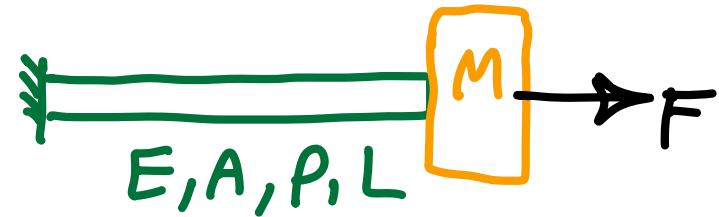
$$E = 200 \text{ GPa} \quad A = 100 \text{ mm}^2 \quad L = 1 \text{ m}$$

Stiffness and extension

$$\sigma_b = E \epsilon = E \frac{\delta}{L}$$

$$\sigma_b = \frac{F}{A}$$

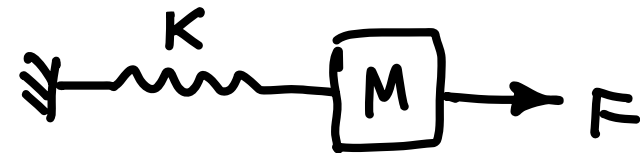
$$F = \frac{EA}{L} \delta$$



$$m_b = \rho A L \ll M$$

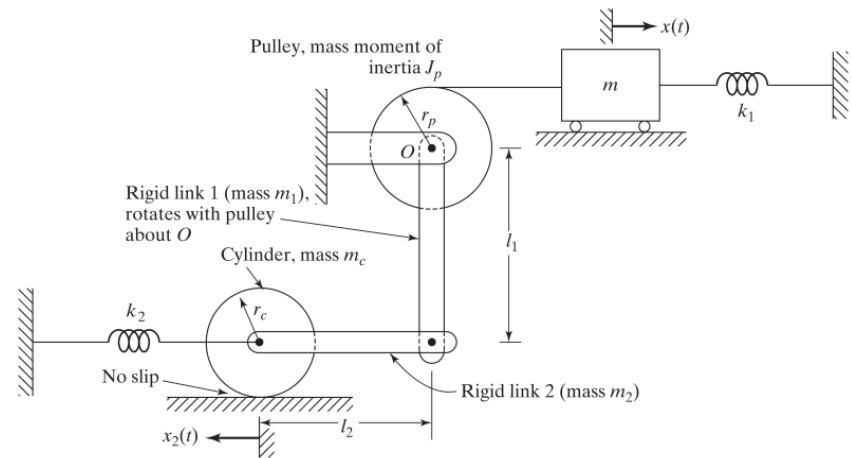
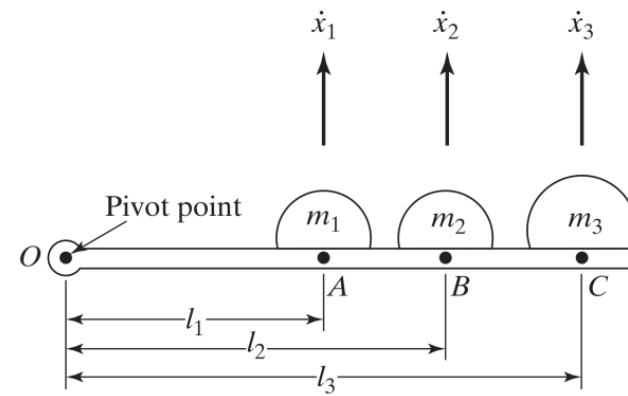
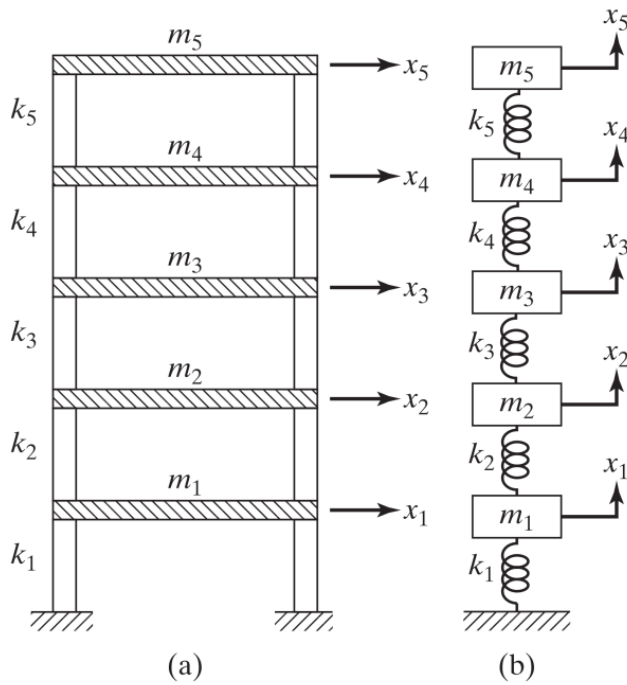
Neglect mass effect of bar.

Neglect stiffness of M .



$$K = \frac{EA}{L} \left(\frac{\frac{\text{N}}{\text{m}^2} \cdot \text{m}^2}{\text{m}} \right) : \left(\frac{\text{N}}{\text{m}} \right)$$

What changes if the rod length doubles?



1.7 Mass or inertia elements

1.7 Mass and rotational inertia

Force and acceleration • Moment and angular acceleration

Translation

$$\sum F_{\text{FBD}} \equiv \sum \mathcal{F}_{\text{k.D.}}$$

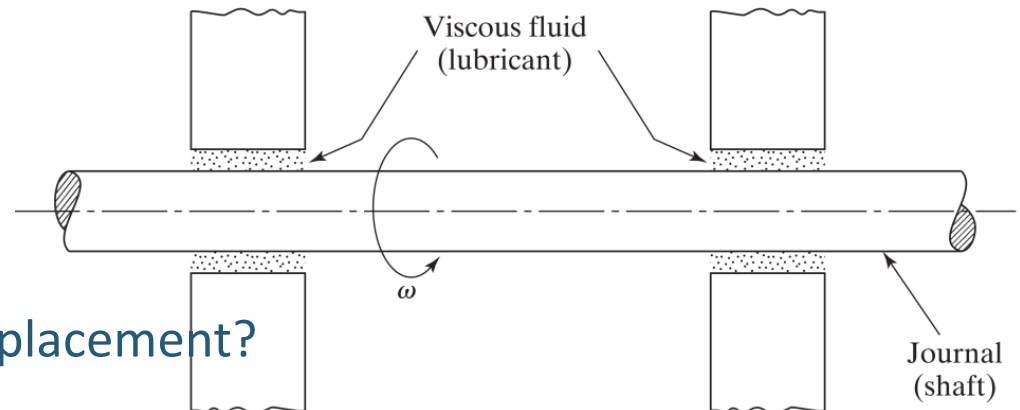
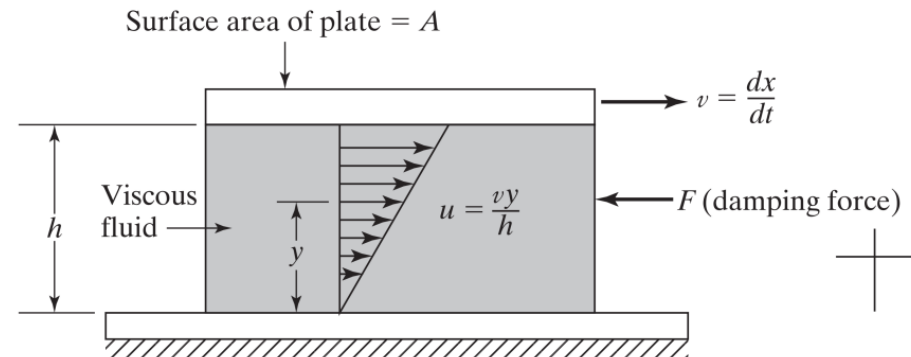
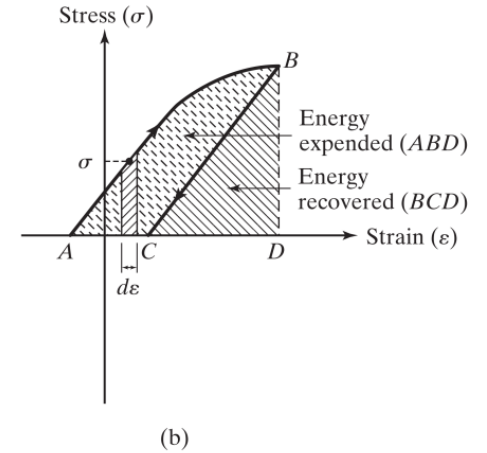
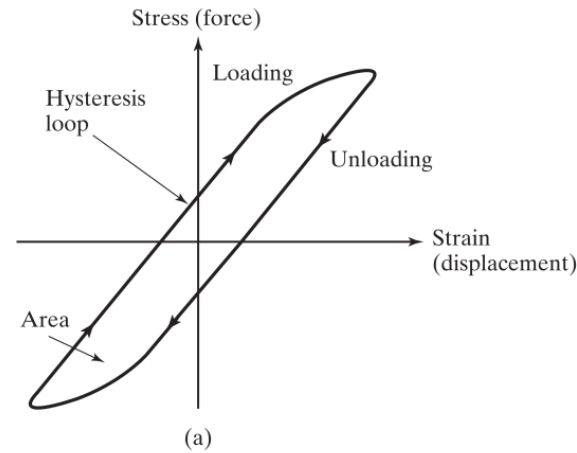
$$\sum \vec{F} = m \vec{a}$$

Rotation about a fixed axis

$$\sum M_{\text{FBD}} \equiv \sum \mathcal{M}_{\text{k.D.}}$$

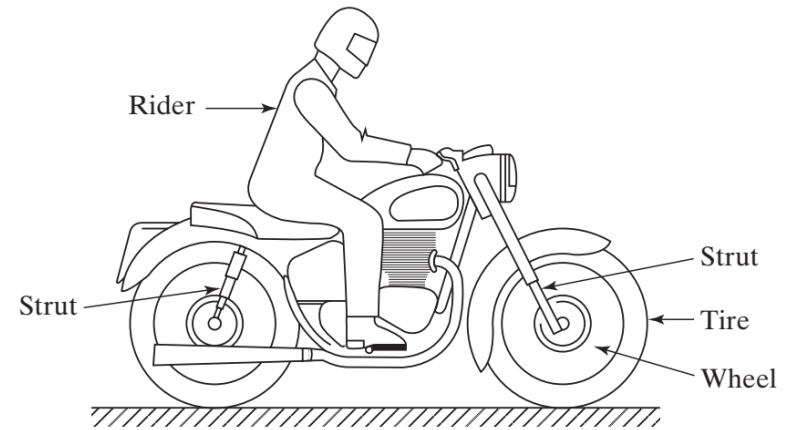
$$\sum \vec{M} = I_o \vec{\alpha}$$

1.8 Damping elements: Viscous, Coulomb, Hysteretic

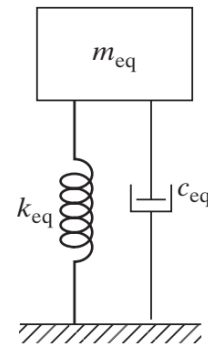


Does damping force always oppose displacement?

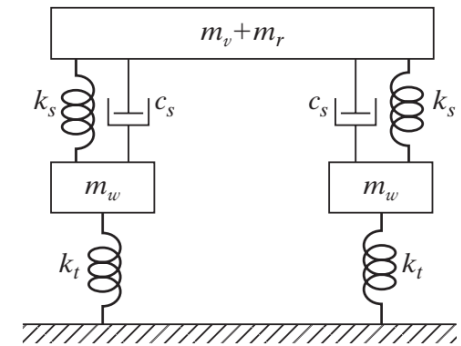
1.9 Physical system to vibration model



(a)



(b)



(c)

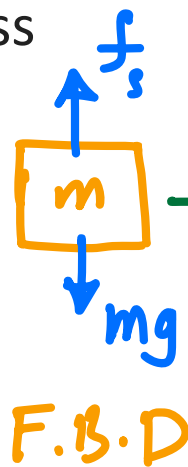
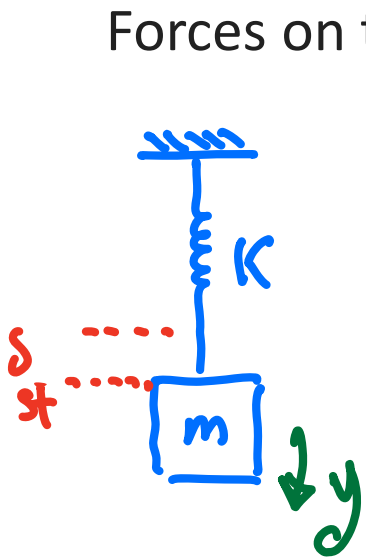
1.9 Physical system to vibration model

Free-body diagram

Positive direction downward; displacement y from the unstretched position

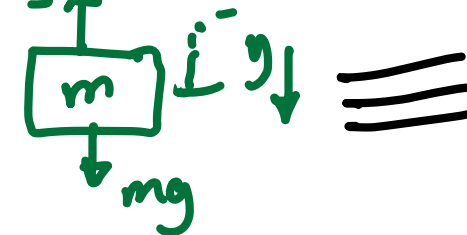
$$+\downarrow mg - f_s - ky \equiv m\ddot{y} \rightarrow \underline{\ddot{y} + \frac{k}{m}y = 0}$$

Forces on the mass

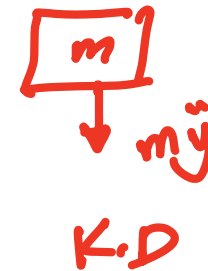


Equilibrium

Newton's second law



F.B.D
when motion



$$\sum F_y = 0 \rightarrow f_s = k\delta_{st} = mg \quad \delta_{st} = \frac{mg}{k}$$

1.9 Physical system to vibration model

Motion about static equilibrium

Static deflection and a new displacement coordinate

Static balance

Coordinate shift

Equation of motion about equilibrium

1.9 Physical system to vibration model

Equation: meaning and checks

Units • Initial conditions • Limiting cases

$$m = 10 \text{ kg} \quad k = 1000 \text{ N/m} \quad c = 20 \text{ N}\cdot\text{s/m}$$

At release: $x = 0.01 \text{ m}$, velocity = 0, $F = 0$

Static deflection? Initial acceleration? Physical direction?

Week 1: closing questions

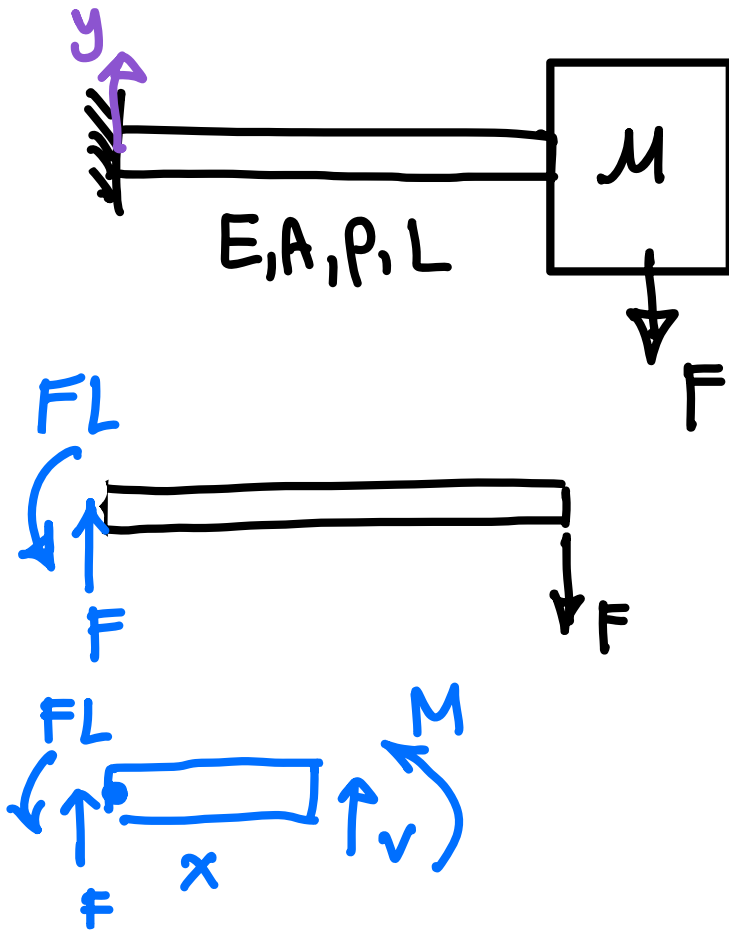
A model should connect the physical system to its motion

What determines the number of degrees of freedom?

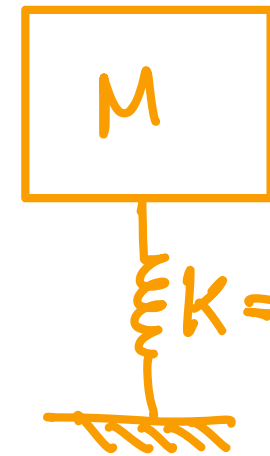
Which elements store energy, and which dissipate it?

Why did gravity disappear from our final equation?

Next week: equivalent elements and harmonic motion



$$M \gg \rho A L$$

$$\equiv$$


$$M = F(x - L)$$

$$M = EI y''$$

$$F(x - L) = EI y''$$

$$EI y' = F\left(\frac{x^2}{2} - Lx\right) + C_1$$

$$EI y = F\left(\frac{x^3}{6} - L\frac{x^2}{2}\right) + C_1 x + C_2$$

$$y(0) = 0, \quad y'(0) = 0$$

$$V = -F$$

$$M + Vx + FL = 0$$

$$y'(0) = 0 \rightarrow C_1 = 0$$

$$y(0) = 0 \rightarrow C_2 = 0$$

$$y = \frac{F}{EI} \left(\frac{x^3}{6} - L \frac{x^2}{2} \right)$$

$$y(L) = \frac{F}{EI} \left(\frac{L^3}{6} - \frac{L^3}{2} \right)$$

$$y(L) = \frac{F}{EI} \left(-\frac{L^3}{3} \right)$$

$$F = K \cdot x$$

$$F = - \left(\frac{3EI}{L^3} \right) y(L)$$

